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BCA II Sem Main/Back Examination 2025-26

BCA–202 : Database Management System

Time : 3 Hours] [Max. Marks : 70

[Min. Passing Marks :

Instructions to Candidates :

Part–A : Short Answer Type Questions (up to 25 words) 10 × 2 = 20 marks. All 10 questions are compulsory.

Part–B : Analytical/Problem Solving questions 5 × 4 = 20 marks. Candidates have to answer 5 questions out of 7.

Part–C : Descriptive/Analytical/Problem Solving questions 3 × 10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form no. 205).

1 _____ **2** _____

Part–A

1. Define Schema.
2. What is a Sub schema ? Give an example.
3. Give an example of a weak entity set with suitable attributes.
4. What is a super key ?
5. What is meant by generalization in E-R model ?
6. Define primary key with suitable example.
7. Define relation and tuple with example.
8. What is a candidate key ?
9. What is a cardinality ratio constraint ? Give example.
10. What is an attribute ?

Part–B [5×4 = 20]

1. What is data independence ? Explain both the categories of data independence.
2. Define database management system (DBMS). State its advantages and disadvantages.
3. What is normalization ? Explain first, second and third normal forms.
4. Explain Entity-Relationship model with ER diagram and suitable example.
5. Write short notes on transaction processing system.
6. Explain with example about concurrent execution of transactions.
7. Write a short note on physical data independence.

Part–C [3×10 = 30]

1. Draw a neat diagram and explain the components of DBMS.
2. Explain the relational database model.
3. Draw an E-R diagram for a college database system and map it into relational database.
4. What is SQL query for the following :
   (i) Generate a list of students whose marks are between 40 and 60.
   (ii) Find average marks.
5. Describe various types of keys with suitable example.

6. To add new field
   (c) To display all records
   (d) Explanation of transaction
7. What is transaction rollback ? (5)